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$$\lim_{x \to 729} \left(\frac{\sqrt[3]{x} - 9}{\sqrt{x} - 27} \right)$$

Neem $t = \sqrt[6]{x}$, dan geldt $x = t^6$.

Dus $\sqrt[3]{x} = t^2$ en $\sqrt{x} = t^3$.

Omdat $x \to 729$, geldt $t \to 3$.

$$= \lim_{t \to 3} \left(\frac{t^2 - 9}{t^3 - 27} \right)$$
$$= \lim_{t \to 3} \left(\frac{(t-3)(t+3)}{(t-3)(t^2+3t+9)} \right)$$
$$= \lim_{t \to 3} \left(\frac{t+3}{t^2+3t+9} \right)$$
$$= \frac{3+3}{3^2 + 3 \cdot 3 + 9}$$
$$= \frac{6}{9+9+9}$$
$$= \frac{6}{27}$$
$$= \frac{2}{9}$$

References